

2022

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

Objective type Questions

1. Choose the correct alternative for each of the following questions : 2×10=20

(a) Which layer of OSI is responsible for hop to hop delivery ?

- (i) Data Link Layer
- (ii) Transport Layer
- (ii) Network Layer
- (iv) None of these

(b) Which layer in OSI takes care of reliability ?

- (i) Data Link Layer
- (ii) Transport
- (iii) Network Layer
- (iv) None of these

(c) Which multiplexing takes care of digital transmission ?

- (i) FDM
- (ii) TDM
- (iii) WDM
- (iv) None

(d) Which line coding scheme is best ?

(i) RZ

~~(ii) NRZ~~

~~(iii) Differential Manchester~~

(iv) Manchester

(e) Which protocol is used for voice over internet ?

~~(i) VOI~~

(ii) SCTP

(iii) TCP

(iv) WDP

(f) Which switching techniques allocates resource before transmitting any data ?

(i) Virtual circuit

~~(ii) Circuit~~

(iii) Datagram

(iv) Message

(g) Which protocol is best in a noiseless channel transmission ?

(i) Stop and wait

(ii) Go Back N ARQ

~~(iii)~~ Selective repeat ARO

(iv) None

(h) MTA stands for :

~~(i)~~ Message transfer agent

(ii) Message travel agent

(iii) Message travel agencies

(iv) Main transfer agent

(i) _____ means loss of energy.

(i) Distortion

(ii) Noise

~~(iii)~~ Attenuation

(iv) None

(j) Which topology follows multipoint connection ?

(i) Star

(ii) Bus

(iii) Mesh

(iv) Ring

Section-B

Short Answer type Questions

Answer any four questions of the following :

5×4=20

2. What are the components of data communication ?

3. Discuss all layers of OSI reference model.

4. What is transmission impairment ? Explain its types.

5. What is IP ? Give the range of IPV₄ classful addressing.
6. Draw the diagram of PCM encoder and explain its functionality.
7. What is switching ? Explain its types.

Section-C

Long Answer type Questions

Answer any two questions of the following :

15×2=30

8. What is selective repeat ARQ ? Draw a diagram of the receiving windows of selective repeat ARQ to explain its functioning.
9. What is congestion ? Explain any five congestion control mechanism.
10. Explain different types of transmission media with diagram.

11. Write short notes on any **three** of the following :

- (a) Remote login
- (b) CSMA
- (c) Hamming code
- (d) Multiplexing
- (e) e-mail

_____ x _____